

**Database Design and Development Individual Coursework
5COSC020W**

Report

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Part A: Database Systems coursework

Question 1: Critical Examination and Analytical Reflection

Critical Evaluation of the EERD in Document A.2:

The EERD presented in Document A.2 offers a foundational attempt to model the Racketminster project. However, it suffers from significant conceptual and semantic issues that undermine its effectiveness in representing the system. Below, I'll outline the key problems with the semantics of the provided diagram:

1. **Ambiguity in Relationships:** The graphic fails to provide a clear definition of relationships between things consistent with Racketminster's real-world setting. The relationship of "Supervises" between the Instructor and the Session, for instance, is not clearly defined. It's not obvious if there may be more than one instructor in a session or just one. This ambiguity causes uncertainty about cardinalities and the exact function of an instructor.
2. **Redundancies and Inappropriate Use of Foreign Keys:** The model's usage of foreign keys (FK) seems excessive at times and repetitious. For instance, the fact that Session has both Instructor_ID and Booking_ID suggests that sessions and bookings are related. But this connection lacks rationale and doesn't represent a true relationship between those things. Either a session should be more closely associated with bookings or it should be connected to an instructor; doing both at once leads to misunderstanding.
3. **Absence of specialisation:** There is a lack of specialisation inside the Session entity for the various session kinds (supervised vs unsupervised). The project brief highlights many session kinds with distinct characteristics, such as group coaching and personal training. An inheritance hierarchy should have been used to capture this in the form of subtypes. The given EERD erroneously represents all sessions as a flat object without differentiation, neglecting to consider these distinctions.
4. **Incorrect Cardinalities:** The cardinalities in the relationships are not well-considered. For example, the "Logged" link between Maintenance_Log and Court uses 1.0...n, which suggests that each log must always involve one or more courts. Since each maintenance log should usually be linked to a specific court, this is incorrect. '1 maintenance log records 0..N courts' is the proper cardinality, indicating that a log may refer to more than one court or none whatsoever. The accuracy and integrity of the data may be affected by these cardinality issues, which might result in inaccurate data constraints across the model.
5. **Uncertain Attribute Placement:** The model isn't always able to distinguish between characteristics that correspond in a relationship and those that belong to entities. Examples of session characteristics are Session_Type and Session_Category, which

are directly related to whether or not the session is monitored. Rather than being straight traits, these characteristics could be more effectively conveyed using a subtype hierarchy.

6. **Absence of Traceability for Block Booking:** The Block_Booking object does not have the required session traceability. Block bookings have their characteristics, but it's unclear how they relate to courts and sessions, or how each session inside a block booking fits into the larger booking process. An important mistake in the conceptual model is the lack of clarity in the relationships.
7. **Mishandling of the Recommendation System:** Although the Recommended_By attribute is present in the Player object, it is not properly represented as a link between players. It should have been a self-referential link between players rather than just a basic feature that allows a player to suggest many other players. The recommended system is not well modelled by the current design.

Reflective Commentary and Approach to Improvement:

I used a methodical approach, concentrating on precisely capturing the real-world semantics of the Racketminster project, to significantly improve the provided design. Key concerns in my design approach included the following:

Entity Identification: Identifying the fundamental entities needed to sustain the system was the first stage in my approach. According to the project brief, these organisations comprised:

- Player
- Booking
- Court
- Park
- Instructor
- Session
- Block_Booking
- Maintenance_Log
- Caretaker

Every entity is essential to the functioning of the Racketminster scheme and represents a substantial part of the system. These entities' associations aid in simulating the interactions between real-world items and activities.

Introducing specialisation: To represent the many features and capabilities, one of the most important enhancements was the introduction of specialisation (or inheritance) in several entities:

- **Instructor:** To reflect the unique duties and responsibilities of each function, I introduced Personal Trainer and Group Coach as specialisations for the Instructor entity.

- **Equipment:** To improve the structure and control of the equipment utilised inside the facilities, I created categories for Moveable Equipment and Fixed Equipment.
- **Court:** For the Court entity, I designated subtypes such as Pickleball Court, Tennis Court, and Multi-Use Court. The various court kinds and their intended purposes are appropriately reflected in this classification.
- **Player:** I made it easier to distinguish between players according to ability level by introducing specialisations for New Players and Experienced Players.
- **Session:** I divided the Session into Supervised and Unsupervised types. To accommodate the various guided sessions, I further specialised under the Supervised table in Fitness Play Coaching, Group Coaching, and Personal Training.

Specialisation enables us to more accurately represent the various session and equipment kinds, resulting in more structured and useful properties like `Session_Type` and `Session_Category` within the context of their parent object.

Clarifying Relationships: A clear and effective conceptual design depends on the relationships between the various things. I made sure that responsibilities and proper cardinalities were assigned to relationships such as "Involves," "Supervises," and "Recommends":

- The word "Supervises" was changed to more accurately describe the work of instructors. Multiple instructors can be present in a session (up to two in the case of group coaching), but each instructor can only be in one session at a time.
- Rather than being a property, the "Recommends" relationship between Player and Player was developed as a recursive relationship. This illustrates how a single player may propose numerous players, clarifying the link.
- Clear links exist between booking courts and sessions to monitor player locations and schedules. To avoid deception, each booking involves a Court and a Player with the appropriate cardinality rules.

Refining Cardinalities: Cardinalities in relationships were something I focused on a lot to make sure they accurately represented the system's practical limitations. As an illustration:

- **Instructor Supervises Session:** One instructor can supervise many sessions, but each session can have only one instructor for supervised sessions.
- **Booking Involves Court:** A court may involve more than one booking at any given moment, but a booking pertains to a single court.
- **Player Recommends Player:** A player may suggest several other players, but each player can only receive one recommendation from other players—or none at all if they weren't suggested.

Block Booking and Traceability: To make sure block bookings could be monitored, I created a connection between `Block_Booking` and `Session`. This facilitates the management of block bookings over time and guarantees the accounting and traceability of each booking session inside a block booking.

Maintenance and Equipment Tracking: By perfecting the Maintenance_Log entity to guarantee proper record-keeping, the design was further enhanced. We can keep track of who is in charge of maintaining the courts and what maintenance tasks have been completed by linking the Maintenance_Log to both the Court and the Caretaker.

Improved Data Integrity: Finally, I verified that each object had an individual primary key {PK} and that foreign keys (FK) are only utilised to support connections when required. Eliminating unnecessary foreign keys and lessening connection ambiguity, this enhanced data integrity.

Conclusion

In conclusion, there were several issues with the original EERD from Document A.2, such as inappropriate cardinalities, missing specialisations, and poor relationship management. Through a thorough examination of the project brief and the actual situation, I was able to make the following major design improvements:

- Introducing specialisation in Sessions, Instructor, Court, Player and Equipment
- Clearly defining relationships and their cardinalities,
- Correctly modelling the recommendation system,
- Ensuring traceability for block bookings,
- Refining the overall structure to better represent the data needs of Racketminster.

Question 2: Table including Entities with brief Explanation

<u>Entity Name</u>	<u>Brief Explanation</u>
w1967351_Block_Booking	Shows that players may reserve numerous unsupervised periods at once. includes properties for booking dates, instructor information, and associated sessions.
w1967351_Booking	Records the specifics of each booking session that players make. Booking ID, date, start time, end time, and associated player and session are examples of attributes.
w1967351_Instructor	Comprises professionals that oversee certain sessions. Name, speciality, and instructor ID are examples of features.
w1967351_Session	Indicates planned playing sessions that can be done under supervision or not. Contains characteristics such as the start and finish times, kind, category, and session ID.
w1967351_Player	Identifies those who are involved in the Racketminster scam. Name, contact details, and Player ID are examples of properties.
w1967351_Park	Symbolises the parks that house the courts. includes information on the park's name, ID, and location.
w1967351_Equipment	Includes the tools, including nets and rackets, that are specific to each court. includes details such as the name, location, associated court, and Equipment ID.
w1967351_Court	Portrays the actual courts that are available for racket sports in parks. Features include court type (pickleball, tennis, or both), location, status, and caretaker associated with the court.
w1967351_Caretaker	Involves those in charge of keeping parks and courts maintained. The Caretaker ID, name, assigned area, and associated maintenance logs are among the attributes.
w1967351_Maintenance_Log	Records-keeping tasks carried out on courts and equipment. includes information such as the maintenance ID, the date, the time, the status, and a summary of the work that was

	done.
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Question 3: Specialisations table with short explanation

<u>General Entity</u>	<u>Specialised Entity</u>	<u>Brief Explanation</u>
w1967351_Session	w1967351_Supervised_Session	An activity where experienced instructors lead the way. Personal training, group coaching, and fitness-play sessions are further categories into which supervised sessions can be divided. Like the amount of teachers and participants, each of these groups has unique qualities and traits.
w1967351_Session	w1967351_Unsupervised_Session	A session that participants may schedule themselves, enabling them to play independently of teacher assistance. For simplicity, players can block-book numerous time slots for these sessions in advance. A session that participants may schedule themselves, enabling them to play independently of teacher assistance. For simplicity, players can block-book numerous time slots for these sessions in advance.
w1967351_Player	w1967351_New_Player	Represents participants in the Racketminster program who have just enrolled. They could benefit from introductory lessons as they might not have much experience with racquet sports. It could be necessary to record other characteristics of this specialism, including preferred learning techniques or first-time courses attended.
w1967351_Player	w1967351_Experienced_Player	Represents athletes with a history of success and expertise in racket sports. Their skill level, prior

		attendance at supervised sessions, and any particular interests in advanced training or competitive play may all be tied to this specialism.
w1967351_Court	w1967351_Tennis_Court	A particular kind of court intended for tennis. The characteristics specific to tennis courts, such as surface type (grass, clay, or hard court) and net height restrictions, would be the emphasis of this specialty.
w1967351_Court	w1967351_Pickleball_Court	A particular kind of court is reserved for pickleball. This specialty would include characteristics unique to pickleball courts, such as the sizes of the Court and equipment configurations.
w1967351_Court	w1967351_Multi-Use_Court	A court is sufficient in size to play pickleball and tennis on. This specialty would have characteristics that show its adaptability to the particular sport being played.
w1967351_Instructor	w1967351_Personal_Trainer	An experienced instructor dedicated to providing customised one-on-one training sessions based on the needs and ability levels of each player. Success rates and personal training specialisations may be included in this entity.
w1967351_Instructor	w1967351_Group_Coach	An instructor in charge of supervising group activities with an emphasis on social play, fitness, or skill development. The group sizes they may oversee and the kinds of group sessions they provide are described in this specialty.
w1967351_Equipment	w1967351_Fixed_Equipment	Court markings and nets are

	t	examples of equipment that is fastened to a court permanently. This speciality could have characteristics that list the dates of installation and upkeep needed.
w1967351_Equipment	w1967351_Moveable_Equipment	Balls, nets, and rackets are examples of equipment that is portable and easily adjustable. Condition tracking and availability status are examples of attributes.

Question 4: Table including Multiplicities and Relationships

<u>Entity Name</u>	<u>Multiplicity</u>	<u>Relationship</u>	<u>Multiplicity</u>	<u>Entity Name</u>	<u>Justification</u>
w1967351_Booking	1	Involves	1	w1967351_player	1. To ensure attendance responsibility, every booking is linked to a single player. 2. A player may be listed on more than one booking, but each booking relates to a single player. 3. This connection makes it possible to monitor each player's involvement in sessions. 4. It facilitates player contact and tailored treatment regarding their reservations.
w1967351_player	1	Recommends	0..n	w1967351_Player	A player can recommend one or more other players. Without offering any recommendations, a player might still exist (zero). A participant may suggest several other people (n).

					This illustrates how one player may have an impact on other players.
w1967351_Booking	1	Involves	0..n	w1967351_Court	Every reservation includes a single court, guaranteeing that each reservation has a precise address. A court's bookings may be zero or more over time, providing flexibility in how the court is facilitated. The scheduling of several bookings for the same court at various times is supported by this connection. It guarantees that every reservation is linked to a particular court, even when courts may be unoccupied at specific times.
w1967351_Booking	1	Includes	1..*	w1967351_Session	- every booking is for

					a single, distinct session. - A session may be linked to more than one reservation. - each booking has to be connected to a session. - Due to session scheduling, every booking is made for a previously planned session.
w1967351_Instructor	1	Supervises	0..n	w1967351_Session	- At least one session is overseen by an instructor. - There may be one instructor monitoring a session or many. - Some sessions do not have optional supervision. The instructors can oversee several sessions.
w1967351_Session	1..*	Consists of	0..1	w1967351_Booking	Flexible scheduling is made possible by the multiplicity

					1..*:0..1, which indicates that a single session may be a part of several block reservations (1..*). On the other hand, a block booking denotes that it may exist independently of any associated session and can be associated with either zero or one session (0..1). This arrangement allows for the flexibility to meet different scheduling requirements while facilitating efficient administration of sessions within block reservations.
w1967351_Park	0..1	Contains	0..n	w1967351_Court	A park may have zero or one caretaker: This statement suggests that a park may exist with just one caretaker or without one.

					A manager can oversee more than one park: which reflects the fact that a caretaker may be in charge of one or several parks, giving park management flexibility. Application in the real world: Describes situations where staffing levels fluctuate, allowing parks that do not always have full-time custodians. Promotes adaptability: By enabling managers to oversee several parks as needed, this multiplicity promotes effective administration.
w1967351_Park	1	maintained by	0..n	w1967351_Caretaker	The multiplicity 1:0..N for the relationship 'Park maintained by Caretaker'

					indicates that each park must have at least one caretaker, while the number of caretakers can vary in situations where parks may exist without a caretaker or have a single caretaker overseeing them. One or more parks might be maintained by a caretaker, giving resource management flexibility. This adaptability guarantees that park managers may allocate caretakers to various parks based on staffing requirements or park usage, while also considering situations where a park may momentarily be without a caretaker because of personnel
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					shortages or demand variations.
w1967351_Court	1	Maintained by	1	w1967351_Caretaker	Every Court Needs a Caretaker: This guarantees that each court has a committed caretaker in charge of its upkeep and maintenance, which is crucial for guaranteeing the courts' quality and safety. A single caretaker is assigned to a single court, guaranteeing targeted supervision and responsibility for that particular establishment . Clear Management System: Because of this diversity, there is a clear management system with defined duties for courts and carers, which

					makes maintenance and monitoring easier. Reflects Realistic Operations: It is reasonable to anticipate that each court at a well-run facility such as Racketminster will have a designated caretaker, which will improve the sports complex's operational effectiveness.
w1967351_Court	1	Has	0..n	w1967351_Equipment	A court may have more than one piece of equipment connected to it, but each court must have at least one piece of equipment available for usage.
w1967351_Maintenance_Log	1	records for	0..n	w1967351_Court	Each maintenance log entry is specific to one court, but a court can have multiple maintenance log entries

					over time.
w1967351_Maintenance_Log	1	records for	0..n	w1967351_Caretaker	A caretaker can have zero or more maintenance log entries over time, but each entry must be linked to a specific caretaker.
w1967351_Session	1	Has	0..n	w1967351_Supervised_Session	Mandatory Supervision: To guarantee quality and safety, each session must have at least one monitored session (1). Flexible Options: Depending on demand, a session may consist of zero or more monitored sessions (0..n). Operational Adaptability: This diversity ensures that sessions can meet the demands of various players by offering flexibility while keeping supervision.

					Efficiency: It permits effective session management by establishing a balance between operational flexibility and safety standards.
w1967351_Session	1	Has	0..n	w1967351_Unsupervised_Session	Session Requirement: To guarantee an organised approach to teaching, each session has to include at least one instance (1). Optional Unsupervised Sessions: To provide participants the freedom to practice on their own, a session may contain zero or more unsupervised sessions (0..n). Training Variety: Players with varying training preferences may be accommodated by this multiplicity,

					which offers both supervised and unsupervised practice. Dynamic Program Design: It improves overall involvement by allowing session organisers to adjust to the demands of the players by including different training styles.
w1967351_Court	1	Includes	0..n	w1967351_Tennis_Court	This multiplicity allows for flexibility in court usage by indicating that a court may or may not be labelled as a tennis court (0..n). Tennis courts can be found in multiples or none at all in any facility, which reflects the dynamic character of sports facilities and their various designs.
w1967351_C	1	Includes	0..n	w1967351_Pi	Every facility

ourt				ckleball_Court	must have a minimum of one pickleball court (1), however, it can have zero to many pickleball courts (0..n), according to the multiplicity 1:0..n. Because of this structure, the layout of the facility may be flexible to accommodate different numbers of pickleball courts according to demand and available space.
w1967351_Court	1	Includes	0..n	w1967351_Multi-Use_Court	Every facility must have one multi-use court (1), however, it can have zero to many multi-use courts (0..n), according to the multiplicity 1:0..n. This provides flexibility depending on the facility's offers and space limits by enabling the potential

					of having additional multi-use courts to serve different sports and events.
w1967351_E quipment	1	type of	0..n	w1967351_Fi xed_Equipm ent	There must be a minimum of one fixed equipment entry (1) for each piece of fixed equipment to be categorised as fixed. The '0..n' designation, however, might result from several instances of fixed equipment, like courts, nets, and seats.
w1967351_E quipment	1	type of	0..n	w1967351_M oveable_Equ ipment	Comparably, every item of moveable equipment needs to be categorised as such to guarantee that each equipment entry has a minimum of one matching type (1). There can be several examples of

					moveable equipment (balls, rackets, and portable goals, for example), which are denoted by '0..n.'
w1967351_Instructor	1	is a	0..n	w1967351_Group_Coach	The instructor has the ability to lead zero or more group coaching sessions, i.e., an instructor can be a group coach (0..n). However, each group coaching session must include at least one teacher (1), demonstrating that a trained instructor is necessary for group coaching.
w1967351_Instructor	1	is a	0..n	w1967351_Personal_Trainer	An instructor may also function as a personal trainer, and they can work with one or more clients one-on-one (0..n). Furthermore, to guarantee responsibility

					and proficiency in the training process, each training session has to have only one instructor (1).
w1967351_Session	1	held in	0..n	w1967351_Park	Every session can only take place in one park (1), however, a park can accommodate zero or more sessions (0..n).
w1967351_Block_Booking	1	belongs to	0..n	w1967351_Booking	1. A single, distinct booking must be connected to every block booking. 2. If a booking is made for a single session, it may not have any block bookings. 3. A booking may be linked to more than one block booking. 4. This illustrates how block bookings can include several sessions in a single

					booking.
w1967351_Player	1	is a	0..1	w1967351_Experienced_Player	Higher skill level players in the sport are identified by the Experienced Player entity. Each player is classified as either an Experienced Player or a New Player, but not both, according to its multiplicity of 1:0..1. Because of this differentiation, coaching techniques, and training chances may be customised to the ability levels of the athletes, guaranteeing that they are provided the right challenges and assistance to improve. Years of experience and specialised abilities are examples of attributes that might improve the training environment.

w1967351_Player	1	is a	0..1	w1967351_New_Player	People who are just getting started in the sport are represented by the New Player entity. Each player is categorised as either an Experienced Player or a New Player, but not both at the same time, according to its multiplicity of 1:0..1. This differentiation facilitates the provision of beginner-specific instruction and resources by coaches and trainers, guaranteeing a nurturing atmosphere for skill development. Characteristics might be age, preliminary skill evaluations, and training objectives, all of which are meant to ease the athlete's transition from less experienced
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					to more experienced stages in their athletic career.
w1967351_Player	1	participates in	0..n	w1967351_Session	1. The number of sessions in which a player may engage is 0...n on the player's side. 2. A minimum of one player (one on the session side) is required for each session. 3. Players might choose to participate in several sessions or none at all thanks to this flexibility. 4. At least one player is required for a session to exist.
w1967351_Booking	1	Involves	1	w1967351_Player	1. To ensure attendance responsibility, every booking must be linked to a single player. 2. The booking mechanism is simple since a player can only have one active booking at a time.

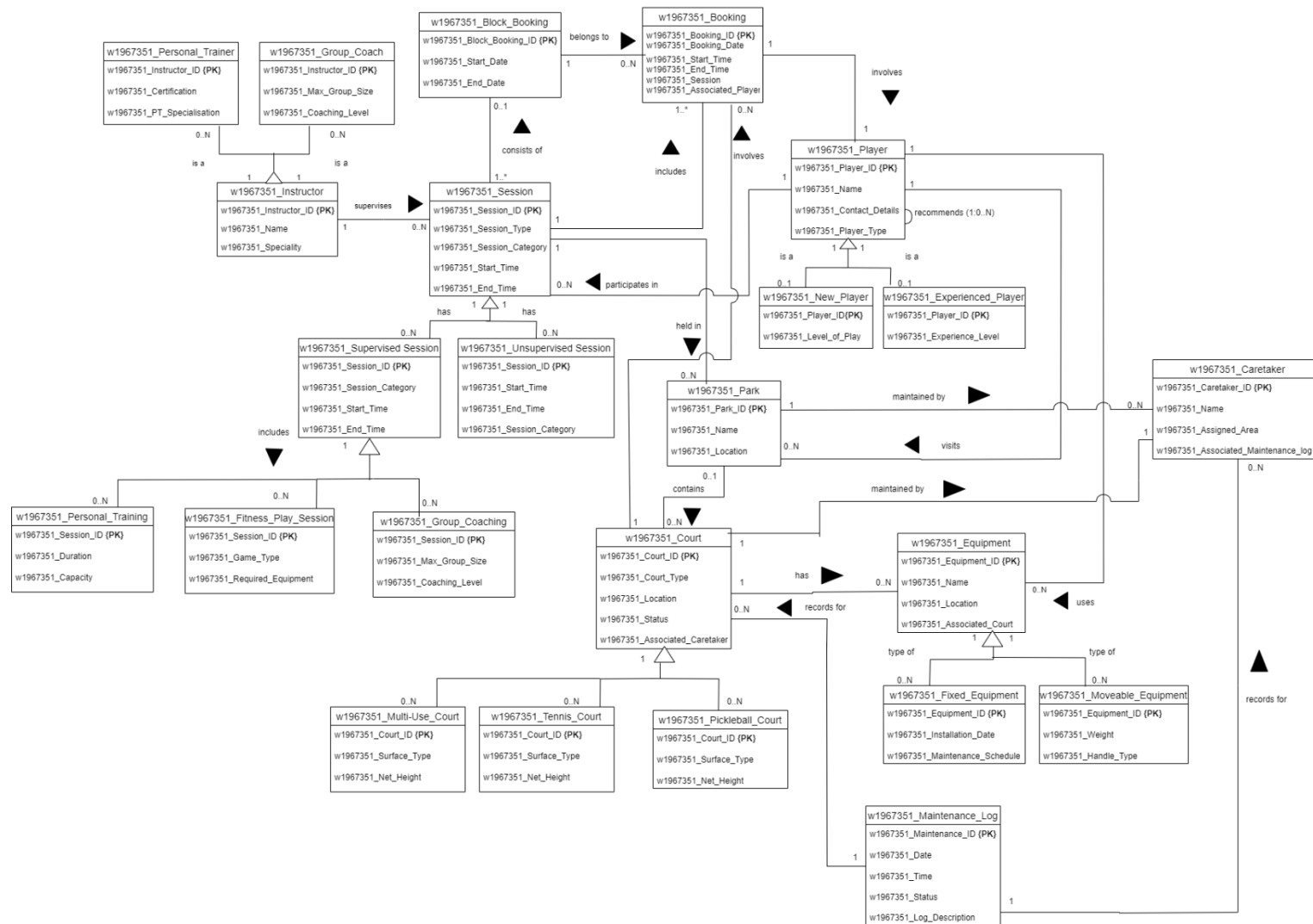
					3. This partnership makes it easier to monitor player activities during a session. 4. It guarantees that, via their lone booking instance, the player's involvement is continuously tracked.
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Question 5: Primary keys with its attributes

<u>Entity name</u>	<u>Attributes for Each Entity (Include PK)</u>	<u>Brief Explanation</u>
w1967351_Player	- w1967351_PlayerID {PK} - w1967351_Name -w1967351_Contact_Details - w1967351_Player_Type	Every player is individually identified by their w1967351_PlayerID primary key. While w1967351_PlayerType indicates if the player is new or experienced, the w1967351_ContactDetails property compiles all contact information.
w1967351_Court	- w1967351_CourtID {PK} - w1967351_Court_Type - w1967351_Location - w1967351_Status - - w1967351_Associated_Car etaker	Each court is included in the Court table, individually recognised by its w1967351_CourtID . Among the attributes are the kind, w1967351_Location , w1967351_Status , and the w1967351_Associated_Car etaker .
w1967351_Session	- w1967351_SessionID {PK} - w1967351_Session_Type - - w1967351_Session_Catego ry - w1967351_Start_Time - w1967351_End_Time	Each session is tracked by the Session database and is uniquely recognised by the w1967351_SessionID . The w1967351_Session_Type , w1967351_Category , w1967351_Start_Time , and w1967351_End_Time are specified by attributes for scheduling purposes.
w1967351_Booking	- w1967351_BookingID {PK} - w1967351_Booking_Date - w1967351_Start_Time - w1967351_End_Time - w1967351_Session - - w1967351_Associated_Play er	Every booking, distinctively identifiable by w1967351_BookingID , is recorded in the Booking table. The w1967351_Booking_Date , w1967351_Start_Time , w1967351_End_Time and w1967351_references to the relevant session and player are examples of attributes.
w1967351_Equipment	- w1967351_EquipmentID	To record information

	{PK} - w1967351_Name - w1967351_Location - w1967351_Associated_Court	regarding the equipment utilised within the facility, I have included this table. Every piece of equipment is individually presentable by its w1967351_EquipmentID , and its characteristics include crucial details such as its w1967351_name , w1967351_location , and w1967351_Associated_Court . This guarantees efficient tracking and administration of equipment resources.
w1967351_Instructor	- w1967351_InstructorID {PK} - w1967351_Name - w1967351_Speciality	The Instructor table provides details about each instructor, uniquely identified by w1967351_InstructorID . Attributes include the instructor's w1967351_name and their w1967351_specialty , allowing for better matching of instructors to sessions and players.
w1967351_Caretaker	- w1967351_CaretakerID {PK} - w1967351_Name - w1967351_Assigned_Area - w1967351_Associated_Maintenance_Log	Each caretaker's primary information is contained in the Caretaker database and is individually recognised by their primary key which is w1967351_CaretakerID . The caretaker's w1967351_name , w1967351_Assigned_area of duty, and w1967351_Associated_Maintenance_Log are provided by attributes so that their actions and responsibilities may be tracked.
w1967351_Maintenance_Log	- w1967351_Maintenance_LogID {PK} - w1967351_Date - w1967351_Time - w1967351_Status - w1967351_Log_Description	The w1967351_Maintenance_LogID uniquely identifies the maintenance actions tracked in the Maintenance Log database. Features include the maintenance activity's w1967351_date and

		w1967351_time , its w1967351_status (finished or pending), and w1967351_description of the work done, offering a thorough record of every maintenance event.
w1967351_Block_Booking	- w1967351_Block_Booking{PK} - w1967351_Start_Date - w1967351_End_Date	The Block Booking table holds details for block bookings. The primary key w1967351_Block_Booking specifically identifies each booking, while the w1967351_start date and w1967351_End_dates specify the duration of the block booking.
w1967351_Park	- w1967351_ParkID {PK} - w1967351_Name - w1967351_Location	The w1967351_Park table holds details for parks. The primary key, w1967351_ParkID , uniquely specifies each park, while the w1967351_Name specifies the name of the park, and w1967351_Location defines the geographic location of the park.

Question 6: Conceptual EERD Notations and Quality**Conceptual EERD for Racketminster**

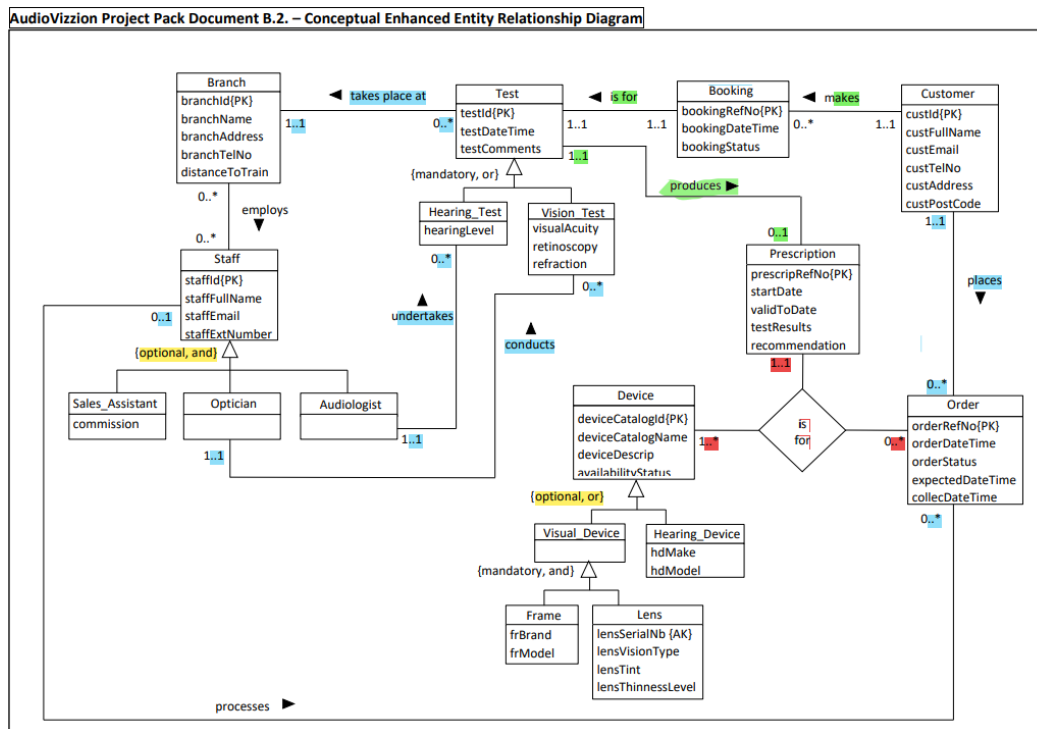
Part B: Database Systems coursework

Question 7: Linking Conceptual EERD to Logical ERD

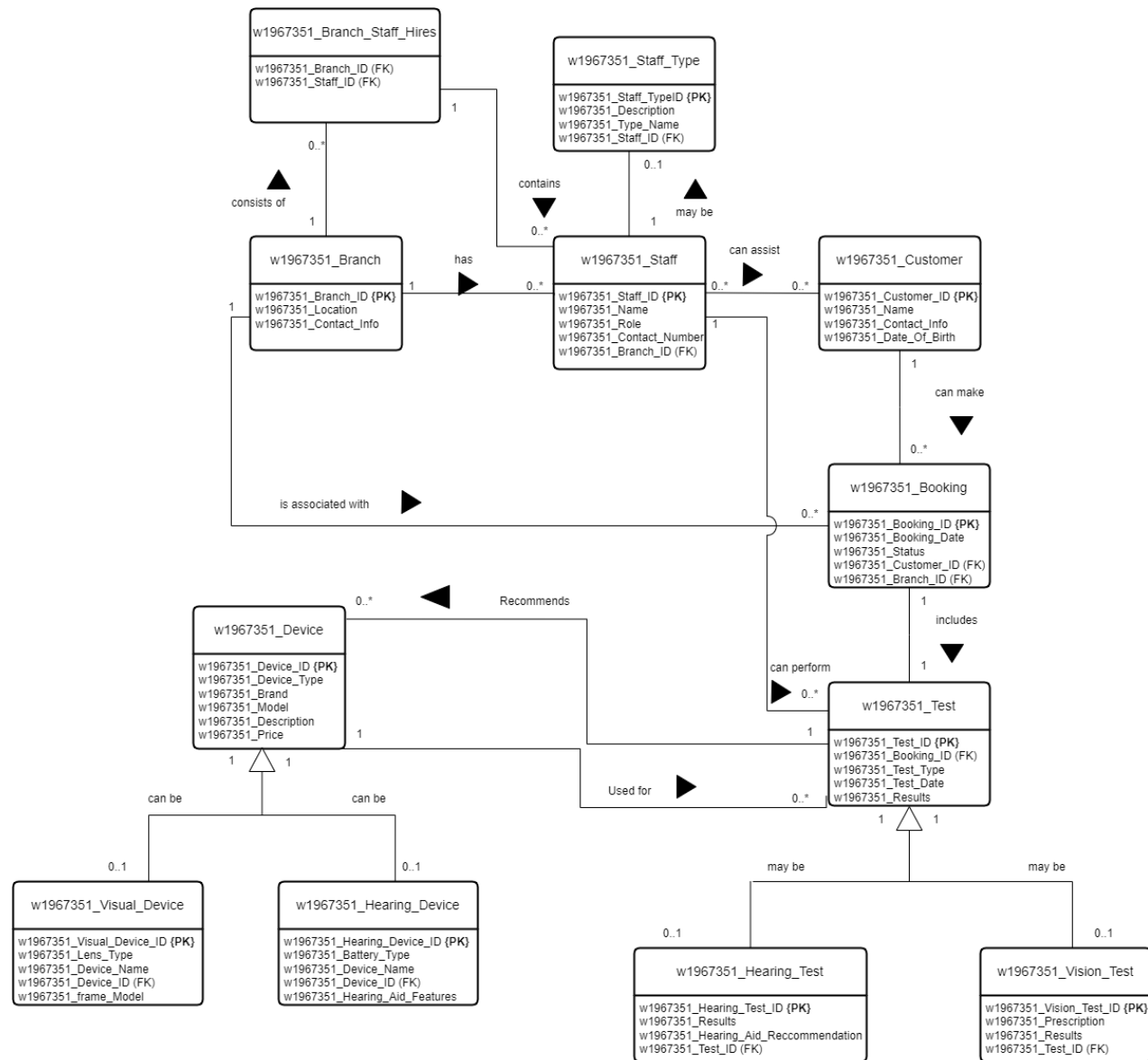
I'll transform the conceptual diagram into a thorough logical Entity Relationship Diagram (ERD) that is especially suited to the AudioVizzion situation and its specifications. This logical ERD will accurately express the structure and connections of entities as described in the conceptual design.

AudioVizzion Project Pack Document B.1. – Simple Project Brief
Author: Francois Roubert & Chathura Sooriya-Arachchi, Board of Managing Directors for AudioVizzion
AudioVizzion is a large retail chain that offers optical and audiology services, and essentially sells spectacles and hearing aids to the public. AudioVizzion operates globally from several branches located all around the United Kingdom. Every branch offers AudioVizzion customers the opportunity to get tested on their vision and/or on their hearing before being able to purchase what the company refers to as Visual Devices (i.e. frames and lenses for glasses) and/or Hearing Devices (i.e. hearing aids).

To create the logical ERD diagram based on the situation, I have mapped all of the significant relationships in the following in accordance with the ten established criteria. To visually differentiate each rule's application, I will also ensure that each one is represented by a specific colour code.



Upon completion, I'll publish the completed ERD diagram below. In order to guarantee precision and clarity, all entities, relationships, and labels will be presented in a way that is comprehensible and accurate.

Logical ERD AudioVizzion

References and resources used

Begg, C. E., and Connolly, T. (2015). Design, Implementation, and Management of Database Systems: A Practical Approach (6th ed., Global Edition). Chapter 1, Chapters 12, 13, and 16 of Pearson Education. [retrieved on 15 October 2024].

Draw.io was used to visualise and layout the diagrams. draw.io

To make sure that the requirements were fully understood, the coursework brief was viewed and examined through the University of Westminster Blackboard platform. [File](#)

"Understanding ERD Diagrams" (2019), an online essay from Lucidchart, is accessible online at <https://www.lucidchart.com/pages/er-diagrams> [retrieved 2nd October, 2024]

"Difference Between Primary Key and Foreign Key" by GeeksforGeeks [Online], accessible at: <https://www.geeksforgeeks.org/difference-between-primary-key-and-foreign-key/> [retrieved at 30th September 2024]

"What Is DBMS's article" Specialisation in DBMS" is accessible online at <https://whatisdbms.com/specialization/> [retrieved October 5th, 2024]

